GEOG 216.01:
FUNDAMENTALS OF GEOGRAPHIC INFORMATION SCIENCE

Department of Geography and Sociology
Samford University
Fall 2026

COURSE INFORMATION

Class Time: Mondays, Wednesdays, and Fridays from 8:00am to 8:55am

Class Location: Ingalls Hall 204 (also known as the GIS Lab)

Instructor: Dr. Jordan Cissell
Phone: 205-726-4766 (office)
Email: jcissel1@samford.edu

(The next-to-last character in my email name is the letter "L", and the last character is the number "1". It's hard to tell them apart when they're printed out side-by-side like that!)

Office Location: Ingalls Hall 242
Office Hours: 9:00am to 11:00am on Mondays and Wednesdays (or by appointment)

Credit Hours: 3

Co-Requisite: GEOG 217: Fundamentals of Geographic Information Science Lab

Prerequisites: None

Textbook: *GIS Fundamentals* (7th ed.) by Paul Bolstad. Eider Press. (2022).

Required Equipment: You will need a USB flash drive or external hard drive with at least 16 GB of storage for saving your GIS data throughout the semester. USB 3.0 is recommended for optimal speed and performance. Note-taking is very important in this class, so you also need to have a class notebook. You are required to bring your USB drive, notebook, and a writing instrument to each class meeting. If you have difficulty acquiring a storage device or notebook, please let me know, and I will be happy to help you figure out a solution.

COURSE DESCRIPTION

GEOG 216 and GEOG 217 are required co-requisites. This course (GEOG 216) is the 3-credit-hour lecture component, and GEOG 217 is the 1-credit-hour lab component.

GEOG 216 emphasizes conceptual material, hands-on demonstrations, and multi-week projects. GEOG 217 emphasizes weekly hands-on lab assignments that apply concepts introduced in lectures and reinforce skills required for completing the multi-week projects. Together, the lecture and lab courses will equip you to be a *geospatial decision maker,* not just a button pusher.

GIS is the most powerful tool in the world for creating, analyzing, and communicating spatial information. Everything is located somewhere, and GIS users can tell you *where* it is, *how* it got there, and *why* its location matters.

The acronym "GIS" has two complementary meanings: geographic information *science* and geographic information *systems*. In this class, we will explore both meanings in tandem toward a foundational understanding of geospatial science and practice. We will explore the fundamentals of GIS as computer-based *systems* for managing spatial data, analyzing spatial relationships, and modeling spatial processes. The productive and ethical use of these systems requires the iterative, reflective approach of the *scientific* method. Therefore, you will also learn how to appraise your objectives, decide which data and analyses are appropriate for your objectives, evaluate your results, and communicate your methods and results effectively.

The skills you will develop in GEOG 216 and GEOG 217 are in higher demand than ever before across a wide range of disciplines, including but not limited to natural sciences, social sciences, business, public health, and journalism. GIS is a cutting-edge science and technology, with exciting new developments occurring every day. This is interesting stuff we are going to be talking about, and right now is an exciting time to be talking about it. I am looking forward to digging into it together this semester!

LEARNING OUTCOMES

After completing this course and GEOG 217, you will be able to:

- Articulate fundamental GIS concepts, interdisciplinary applications of spatial technology, and the value of a geographic perspective in addressing real-world issues.
- Use scientific methods to solve applied problems through laboratory and other active investigative work.
- Locate, acquire, evaluate, create, edit, and display spatial datasets and metadata.
- Differentiate and apply vector and raster data models, discrete and continuous datasets, and data measurement scales.
- Recognize properties of map scales, projections, and coordinate systems and address challenges associated with their use.
- Select and use geoprocessing techniques for data management and basic spatial analysis.

- Communicate results of spatial data analysis by producing maps and other geovisualizations.

COURSE COMMUNICATIONS

Throughout the semester, I will make all course materials and info available to you on Canvas. Whenever I communicate new updates to the entire class, I will do so using the Canvas Announcements tab. I will use Samford email and/or Canvas Inbox for individual communications as need be.

Please feel free to email me at my Samford email address or Canvas Inbox at any time. I will do my best to respond by the end of the next business day.

OFFICE HOURS

I am generally in my office whenever I am not in the classroom, but the office hours listed above are times that I am guaranteed to be in my office and available to meet. You are always welcome to drop by and ask me a question, to talk about geography, or anything else!

If you are not able to meet during my listed office hours, no worries. Please email me to set up a different time for us to meet. You are also always welcome to come up and talk with me in the classroom before or after class.

ATTENDANCE POLICY

In-person attendance at every class meeting is expected and required, and I will take attendance during each class meeting. **If you have more than 6 unexcused absences, you will receive a grade of FA for the course.** (I.e., if you have 6 unexcused absences, there will be no penalty. On the 7th unexcused absence, you will receive a grade of FA.)

- If you miss a class meeting because of illness, a medical or family emergency, exceptional hardship beyond your control, or Samford-related travel, *and* you provide me with official documentation of your circumstances within 2 business days of your absence (or before 11:59pm on our course's Final Presentations day, whichever comes first), your absence will be eligible to be counted as excused, at my discretion.

- I am not authorized to excuse long-term (i.e., more than three class meetings in a row) or recurring (e.g., frequent absences throughout the semester due to a chronic health issue) absences from class. If you need to have long-term and/or recurring absences this semester, please let me know, and I will help you get in

touch with the Academic Success Center, the CARE Team, and/or the Office of Accessibility and Accommodations (as applicable). They will discuss your situation with you and make the final determination on whether your absences should be counted as excused or unexcused.

An excused absence does not relieve you of responsibility for the academic work and/or materials from the class meeting that you missed. However, you will not be penalized for an excused absence, and you will be given the opportunity to make up missed work. Please see the POLICY REGARDING MAKEUP ASSIGNMENTS section later in this syllabus for more information about the process for making up missed exams and assignments.

COURSE ASSIGNMENTS

1. **Projects (2):** You will complete 2 Projects throughout the semester, in which you will integrate and apply multiple GIS techniques and concepts toward real-world questions and problems. The deliverable for each Project will be a report. In your reports, you will be expected to not only communicate your GIS methods and results, but also to explain the decision-making process behind your methods, support your decisions with insights from peer-reviewed literature, and interpret the meaning of your results. We will discuss more details about the Project reports during the semester, and I will provide rubrics in Canvas.

   The due date for each Project is listed in the PROVISIONAL COURSE SCHEDULE. The Projects are due by 11:59pm on the Friday of the week for which they are listed.

2. **Final Project:** The Final Project is the culminating activity of your time in GEOG 216. The Final Project will require you to address a spatial objective/question of your choice using GIS methods that you determine and GIS data that you collect and/or create. Your Final Project topic must be approved by me by an assigned time. As with the other Projects described above, you will communicate your process, justify your decisions, and interpret your results. The Final Project is more in-depth than the other Projects and will consist of the following 3 deliverables:

   a. Final Project Proposal
   b. Final Project Report
   c. Final Project Presentation

   All 3 deliverables will factor into your Final Project grade. We will discuss more details about the Final Project during the semester, and I will provide rubric(s) in Canvas.

   The due date for each Final Project deliverable is listed in the PROVISIONAL COURSE SCHEDULE. The Final Project Proposal is due by 11:59pm on the Friday

of the week for which it is listed. The Final Project Report is due by 11:59pm on the day listed. Final Project Presentations will take place during the time listed.

3. **Esri Academy Courses (3):** You will complete 3 Esri Academy Courses throughout the semester. These courses are hands-on tutorials created by Esri (the company that makes ArcGIS Pro and ArcGIS Online). They will help you further develop your GIS skills. We will discuss more details about the Esri Academy Courses during the semester.

   The due date for each Esri Academy Course is listed in the PROVISIONAL COURSE SCHEDULE. Each Esri Academy Course is due by 11:59pm on the Friday of the week for which it is listed.

4. **Portfolio:** At the end of the semester, you will compile your Projects, Final Project, Esri Academy Course certificates, and other materials in a digital Portfolio. This Portfolio will be both 1) a handy reference that you can consult when doing GIS analysis in the future and 2) documentation of your GIS skills that you can present to future employers. We will discuss more details about the Portfolio during the semester.

   The Portfolio is due by 11:59pm on the date listed in the PROVISIONAL COURSE SCHEDULE.

GRADE BREAKDOWN

Project 1	25 %
Project 2	25 %
Final Project	35 %
Esri Academy Courses	5 %
Portfolio	10 %
	100 %

A	93 – 100.0 %
A -	90 – 92.9 %
B +	87 – 89.9 %
B	83 – 86.9 %
B -	80 – 82.9 %
C +	77 – 79.9 %
C	73 – 76.9 %
C -	70 – 72.9 %
D +	67 – 69.9 %
D	63 – 66.9 %
D -	60 – 62.9 %
F	< 60 %

Note: I generally do not provide extra credit assignments on an individual basis. If an extra credit opportunity arises during the semester, it will be made available to *all* students, and I will communicate it to the entire class.

POLICY REGARDING MAKEUP ASSIGNMENTS

If you miss a Project, Final Project deliverable, Esri Academy Course, or Portfolio due date because of an excused absence (please see the definition of an excused absence in the ATTENDANCE POLICY section of this syllabus), then we will work together to schedule a new time for you to complete the assignment.

If you miss the original due date for a Project, Final Project deliverable, or the Portfolio **without** a documented excuse, you will receive a grade of 0 for the assignment.

If you miss the original due date for an Esri Academy Course **without** a documented excuse, 10 points (out of 100) will be deducted from the maximum possible grade for each day that the assignment is late. For example, if Esri Academy Course 1 is due Friday, September 4, and you submit Esri Academy Course 1 on Saturday, September 5, then the maximum possible grade you can earn for Esri Academy Course 1 is a 90. If you submit Esri Academy Course 1 on Sunday, September 6, then the maximum possible grade you can earn is an 80. And so on. Esri Academy Course submissions without a documented excuse will **not** be accepted after 11:59pm on our course's Final Presentations day.

I understand that emergencies have a knack for popping up when we least expect them, and I am willing to work with you on scheduling makeup opportunities so long as you provide documentation within 2 business days of your absence (or before 11:59pm on our course's Final Presentations day, whichever comes first). If you know beforehand that you will miss an assignment due date because of an excused absence, please let me know ahead of time, either in person or via email. The earlier we can discuss the situation, the sooner we can make the proper arrangements to get you caught back up.

NOTICE OF ARCGIS ONLINE ACCOUNT EXPIRATION

Throughout this semester, you will access ArcGIS Online and ArcGIS Pro through the Samford Geospatial Portal using your Samford ID. One year after you graduate, your Samford ArcGIS Online account will be disabled, and you will no longer be able to access any content stored in your Samford ArcGIS Online account. If you wish to keep any content from your Samford ArcGIS Online account after you graduate, you can create a new ArcGIS Online account and transfer your content to the new account. (Esri has several options available for individual users to create ArcGIS Online accounts, including ArcGIS for Personal Use or Public Account options.)

USE OF TUTORS AND/OR AI FOR ASSIGNMENTS

You are permitted to seek assistance from tutors and/or artificial intelligence (AI) for this course. However, it is imperative that all submitted work is completed by **you**. Tutors or generative AI tools can provide guidance and support, but they **must not** be used to complete assignments on your behalf. If you choose to get help from tutors and/or AI in this course, please remember the following:

- All submitted work must be the product of your own effort, thought, and understanding. A tutor and/or AI can help clarify concepts or provide guidance, but the actual work must be done by you.

- Assignments and projects should accurately reflect the concepts and methods discussed in class and/or outlined in the assignment instructions. This ensures that you are applying what you have learned in class. Exploration of different methods is not discouraged, but if you choose to use them, you may be asked to demonstrate your abilities.

- Samford Geography faculty are experienced GIS users and teachers, and we are adept at recognizing when work does not align with a student's usual performance or appears to be completed by someone else. Any work suspect of not being your own will be subject to further review and potential academic consequences.

By adhering to these guidelines, you will gain a deeper understanding of GIS and develop the skills necessary for success in this field.

EXPECTATIONS AND REQUIREMENTS

- Food and open drink containers are not allowed in the GIS Lab.

- Professional conduct is expected and required. Please pay attention, ask questions when you have them, and stay on task. I know that computers can present a wide range of distractions. Please resist the temptation to check email, scroll social media, or shop online during class time, or you risk falling very far behind on course material.

- Please arrive on time for every class meeting and stay for the entire length of every class meeting. This is expected even when I am not lecturing. Much of our time together will be spent doing hands-on demonstrations and/or in-class GIS work. Active and full participation in these activities is essential to your success in the class and your development as a geospatial decision maker.

- Please expect to spend additional time outside of class in the GIS Lab working to complete assignments.

Potential for Transition to Online or Remote Course Delivery

Students must attend class and check email and Canvas regularly for any important updates from the course instructor and/or university administration in the event that the course needs to transition to an online and/or remote delivery format.

Technology Requirements

Because of the possibility of needing to participate in the course virtually at some point during the semester, Samford University requires that all students have the following technology available to them throughout the semester:

- A Windows 11 or Macintosh (macOS version 14 or higher) desktop or laptop computer. (Although Chromebooks and mobile devices are capable of accessing Canvas, they may not be compatible with certain third-party software.)
- A web cam. (These are built into most modern laptops.)
- A microphone and speakers, headphones, or earbuds. (These are also built into most modern laptops.)
- The most recent version of the Chrome or Firefox web browser. (The Safari browser sometimes has problems with accessing online courses.)
- A stable Internet connection capable of at least 2 Mbps upstream and 5Mpbs downstream. (You can test the speed of your connection at http://speedtest.net)

If at any point in the semester you have difficulty acquiring or accessing these supplies, please let me know, and I will be happy to work with you to figure out a solution.

Emergency Readiness

RAVE is the primary method of communication used by Samford University during a campus emergency. If you have not registered for RAVE Alerts, please go to mySamford at https://my.samford.edu, then click on Quick Links and My RAVE Emergency Alert Info to update your RAVE Emergency Alert Information. Note that the My RAVE Emergency Alert Information link is also accessible on mySamford under Tools, My Personal Information and Tools, Emergency Resources.

Samford University utilizes Alert Samford for desktop, laptop, tablet, and mobile devices to provide students with information, procedures, and links about what to do in the event of a variety of emergency situations that could occur on our campus. If you do not already have the Alert Samford app on your mobile device, laptop, desktop, or tablet, please go to mySamford at https://my.samford.edu and navigate to Tools, Emergency Resources, Alert Samford: Get the App for instructions on downloading the App. Once

you have downloaded the App, please take time to review the information provided. It is important that you know what to do in case of a campus emergency.

AMERICANS WITH DISABILITIES ACT

If you require accommodations for this course due to disability or chronic health conditions (medical, psychological, mobility, or sensory), then I am happy to work with you! First, please complete the request process with the Office of Accessibility and Accommodations (OAA). Once eligibility is determined by OAA, please send me official notification of your accommodation eligibility through the OAA online portal. Please also schedule a meeting with me to discuss, because there may be additional steps necessary to begin using accommodations in the course. Required steps may include you scheduling with the OAA Testing Center, signing a recording agreement, completing an attendance modification form, and more. Once we have discussed all the details, we will work together to implement accommodations. Accommodations may not be applied retroactively, and requests made with insufficient notice are typically not possible. If you need assistance requesting, using, or modifying your accommodations in this course, please contact access@samford.edu or visit www.samford.edu/dr for more information. The OAA office is located in DBH 203.

TITLE IX

Samford University and its faculty are committed to creating and maintaining a safe learning environment for all students and the entire University community. If you or someone you know has experienced sexual harassment, sexual assault, relationship violence, stalking, or discrimination based on sex or gender, please know that help and support are available. Samford University strongly encourages all community members to report incidents of sexual misconduct to the Title IX Office. You may contact the Title IX Office at 205-726-2736 or titleix@samford.edu.

Please be aware that all faculty members are required to disclose information concerning suspected or alleged sexual harassment or other violations of the Samford University Sexual Misconduct Policy to the Title IX Office. You can also make a disclosure yourself, including an anonymous report, through the Sexual Misconduct Report Form accessible on the Title IX website at https://www.samford.edu/students/title-ix/. If the Title IX Office receives information about an incident, they will reach out to offer information about resources, rights, and available procedural options. It is a student's choice whether they wish to engage or respond to that outreach.

If you or someone you know has experienced sexual harassment, sexual assault, or any form of sex-based discrimination, you are encouraged to report it to the Title IX Office. The Title IX Office is here to help ensure your safety, provide supportive measures, and explain your options.

If you would prefer to first speak with someone confidentially—before making a report—you may contact a confidential resource. A full list of confidential resources is available in the online student handbook. You can also reach out to the Counseling Office at 205-726-4083 or the Office of Spiritual Life at 205-726-2825.

ACADEMIC INTEGRITY

The University catalog provides that "students, upon enrollment, enter into voluntary association with Samford University. They must be willing to observe high standards of intellectual integrity; they must respect knowledge and practice academic honesty." Unless an instructor provides specific instructions to the contrary, students are expected to conduct their course work independently. In no event may students cheat, copy, or plagiarize the work of others. The use of artificial intelligence tools (AI) in this course is subject to approval of the course instructor. When drawing from various resources for assignments, students shall provide proper citations, footnotes, and bibliographic information.

The instructor of this course reserves the right to issue an academic penalty to students who are found responsible for violations of Samford's honor code (undergraduate programs) or academic integrity (graduate and professional programs). Academic penalties assigned by the instructor may include, but are not limited to, a grade of zero for the assignment/assessment and/or a lowered cumulative course grade. Academic penalties issued by the instructor are separate and apart from any sanction issued by the honor council.

Academic integrity violations shall be governed by the policy that applies in the school or program in whose course or activity the alleged violation occurred.

Please refer to the Samford University Student Handbook for additional details.

Please do not engage in any acts that either constitute or resemble academic misconduct while enrolled in this course or in any other course during your time at Samford University. If you are ever unsure about any of the material presented in this course, please feel free to ask me about it, either in class, via email, or during office hours. I am always happy to help with any questions you may have! We are going to be talking about a lot of interesting stuff in this class, and I would much rather you ask me and your classmates questions and truly learn the material than copy someone else's work and miss out on all of the cool stuff that geography has to offer.

STUDENT GOVERNMENT ASSOCIATION
SAMFORD UNIVERSITY HONOR PLEDGE

Preamble

The Samford University motto – *For God, For Learning, Forever* – brings together our commitment to Christian heritage, the pursuit and transmission of knowledge, and enduring virtues. Only a particular kind of person and institution can aspire to and achieve such demanding principles. The various members of the Samford University community must embrace honor in all areas of life and scholarship. An honor pledge is therefore an appropriate expression of this commitment.

Honor Pledge

I believe that the members of the Samford community possess dignity and are worthy of honor. I understand that honoring others fosters academic achievement, personal growth, and spiritual development. I commit to hold myself and my peers to the standards of conduct and academic integrity maintained by the University.

To this I, _____, pledge my sacred honor.

For God, For Learning, Forever

PROVISIONAL COURSE SCHEDULE

Please note that this schedule is provisional, and I reserve the right to alter the schedule throughout the semester if the need arises. However, if I do alter the schedule in any way, I promise to let you know ahead of time via Canvas Announcements and to post an updated version of the syllabus in Canvas.

In-person attendance at every class meeting is expected and required, and I will take attendance during each class meeting.

Week #	Day	Date	Topic	Reading	Assignment
1	M	Aug. 24	Introduction to the Course	-	-
	W	Aug. 26	GIS: System and Science		
	F	Aug. 28			
2	M	Aug. 31	Interacting with GIS and GIS Data	Ch.1 (pg. 1-15)	**Esri Course 1**
	W	Sep. 2			
	F	Sep. 4			
3	M	Sep. 7	**NO CLASS – Labor Day**	Ch. 2 (pg. 25-30, 36-45, 49-58, 72-73)	-
	W	Sep. 9	Interacting with GIS and GIS Data		
	F	Sep. 11			
4	M	Sep. 14	Acquiring and Managing GIS Data	Chapter 7	-
	W	Sep. 16			
	F	Sep. 18			
5	M	Sep. 21	Cartography	Ch. 4 (pg. 168-175) Ch. 9 (pg. 375-382)	-
	W	Sep. 23			
	F	Sep. 25			
6	M	Sep. 28	Coordinate Systems	Chapter 3	**Project 1**
	W	Sep. 30			
	F	Oct. 2			
7	M	Oct. 5	Creating and Editing GIS Data	Ch. 4 (pg. 147-167, 176-185) Ch. 5 (pg. 193-203) Ch. 9 (pg. 416-417)	-
	W	Oct. 7			
	F	Oct. 9			
8	M	Oct. 12	**NO CLASS – Fall Break**	Ch. 8 (pg. 335-339) Ch. 9 (pg. 366-375)	**Esri Course 2**
	W	Oct. 14	Queries		
	F	Oct. 16			
9	M	Oct. 19	Joins and Overlays	Ch. 8 (pg. 340-347) Ch. 9 (pg. 363-365, 384-409)	-
	W	Oct. 21			
	F	Oct. 23			
10	M	Oct. 26	Raster Analysis	Ch. 10 (pg. 435-439) Ch. 11 (pg. 473-481, 495-496)	-
	W	Oct. 28			
	F	Oct. 30			
11	M	Nov. 2	Web-Based GIS and Sharing GIS Data	Ch. 15 (pg. 642-645)	**Project 2**
	W	Nov. 4			
	F	Nov. 6			
12	M	Nov. 9	Remote Sensing	Ch. 6 (pg. 239-244, 266-279)	**Final Project Proposal**
	W	Nov. 11			
	F	Nov. 13			
13	M	Nov. 16	In-Class Final Project Analysis	-	**Esri Course 3**
	W	Nov. 18			
	F	Nov. 20			

14	M	Nov. 23	In-Class Final Project Analysis	-	-
	W	Nov. 25	**NO CLASS - Thanksgiving**		
	F	Nov. 27			
15	M	Nov. 30	In-Class Final Project Analysis	-	-
	W	Dec. 2			
	F	Dec. 4	**NO CLASS – Study Day**		
16	M	Dec. 7	**Final Project Presentations (8:00am – 10:00am)**	**-**	• **Final Project Report** • **Portfolio**